Question-1

Peas

- Performance measure:

The robot should deliver items on time, avoid accidents and deliver the correct items to the right people.

- Environment:

The robot operates around university buildings, classrooms, campus roads and students.

- Actuators:

The robot uses wheels to move, robotic compartments to carry items and speckers to communicate with people.

- Sensors:

The robot uses camaras to identify obstacles, GPS

to find its location and ultra-sonic sensors to detect nearby objects.

Question-2

- **Chess → Fully observable**
  Both players can see all pieces and their positions on chessboard.
- **Taxi driving → Stochastic**
  The outcome is uncertain because of traffic, pedestrians and unpredictable drivers behaviour.
- **Crossword puzzle → Static**
  The puzzle remains unchanged while the agent is solving it.
- **Chess → Discrete**
  Chess has finite number of squares, pieces and legal moves.
- **Video game (Multiplayer) → Multi-agent**
  Multiple players interact and compete to achieve their goal.

- Medical diagnosis system → Partially observable
The system can't directly observe every internal condition of the patient and must use symptoms and test results to make decisions.

Question-3

- BFS
  - Completeness: Complete if the branching factor is finite
  - Optimal: Optimal when all step costs are equal
  - Time Complexity: $O(b^{d+1})$
  - Space Complexity: $O(b^d)$

- DFS
  - Completeness: Not always complete. It may get stuck in an infinite path
  - Optimal: Not optimal. It may

not find the shortest path.

- Time Complexity: $O(b^m)$
- Space Complexity: $O(mb)$

. b = branching factor (no. of successors/Node)
. d = depth of shallowest goal
. m = max depth of search Node

Question-4

. DLS:

Depth Limited Search is a type of DFS that sets a maximum depth limit to prevent searching too deeply.

. Limitations

- It may miss the goal if it is beyond the depth limit.
- It is incomplete DFS

. IDS:

Iterative deepening

Search repeatedly runs DLS increasing the depth limit each time (0, 1, 2, 3, ...). It overcomes DLS's limitation by eventually reaching deeper goals while using Less memory than BFS.

It is complete when the branching factor is finite and optimal when all step costs are equal.

Question-5

Step	Expand	Explore	Frontier
0	ϕ	ϕ	S:0
1	S(0)	[illegible] S	A:1, C:3, B:4
2	A(1)	S, A	C:3, D:3, B:4, E:6
3	C(3)	S, A, C	D:3, B:4, J:5, E:6, K:9
4	D(3)	S, A, C, D	B:4, I:4, J:5, E:6, H:7, K:9
5	B(4)	S, A, C, D, B	I:4, F:5, J:5, E:6, H:7, K:9, G:11
6	I(4)	S, A, C, D, B, I	F:5, J:5, E:6, H:7, K:9, G:11
7	F(5)	S, A, C, D, B, I, F	J:5, E:6, H:7, K:9, G:11
8	J(5)	S, A, C, D, B, I, F, J	E:6, H:7, K:9, G:11
9	E(6)	S, A, C, D, B, I, F, J, E	H:7, K:9, G:11
10	H(7)	S, A, C, D, B, I, F, J, E, H	K:9, G:11
11	K(9)	S, A, C, D, B, I, F, J, E, H, K	G:11
12	G(11)		Goal reached

- **UCS**
  - Traversal : S, A, C, D, B, I, F, J, E, H, K, G
  - Optimal path: S → B → G
  - Total path cost: 4 + 7 = 11
- **BFS**
  - Traversal : S, A, B, C, D, E, F, G
  - Solution path: S → B → G

Question-6

Step	Expand	Explore	Frontier
0	ϕ	ϕ	ϕ S
1	S	{S}	[A, B, C]
2	A	{S, A}	[B, C, E]
3	B	{S, A, B}	[C, E, D]
4	C	{S, A, B, C}	[E, D]
5	E	{S, A, B, C, E}	[D, F]
6	D	{S, A, B, C, E, D}	[F]
7	F	{S, A, B, C, E, D, F}	[G]
8	G	,	Goal reached

- Traversal : S → A → B → C → E → D → F → G
- Solution path: S → A → E → F → G
  (4 Edges)

Question-7

IDS:

1. Iterative deepening search repeadly performs Depth-Limited search increasing the depth limit by 1 each time

until the goal is found.

2. IDS combines the low memory usage of DFS with the completeness and shallowest goal finding ability of BFS.

3. Yes, IDS is optimal when all steps costs are equal, because it finds the goal at the shallowest depth first.

Question-8

Algo	Complete	Optimal	TC	SC
DFS	No (Yes in finite graph)	No	$O(b^m)$	$O(bm)$
BFS	Yes	Yes, if step costs are uniform	$O(b^{d+1})$	$O(b^d)$
UCS	Yes, if every step cost $\geq \varepsilon > 0$	Yes	$O(b^{1+\lfloor C^*/\varepsilon \rfloor})$	$O(b^{1+\lfloor C^*/\varepsilon \rfloor})$
IDDFS	Yes	Yes, if step cost are uniform	$O(b^d)$	$O(bd$